Nonclosedness of an ordinary Sobolev restriction of a magnetic Airy operator

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Abstract

An ordinary Sobolev domain need not give a closed realization of a magnetic differential expression on an unbounded set. For a half-plane magnetic complex Airy expression, we construct an explicit graph limit showing that the restriction printed with the ordinary H_0^1 domain is nonclosed for every fixed real parameter. Under the bounded-inverse definition of the resolvent, its spectrum is all of $\mathbb{C}$; if spectrum is reserved for closed operators, the domain fails the prerequisite. This domain correction leaves the all-parameter spectral question for the natural closed magnetic Dirichlet realization unanswered.

Almog's half-plane magnetic Airy problem [1] is printed with an ordinary H_0^1 domain. The spectral problem in Almog–Helffer [2, pp. 1–2 and 6] concerns a closed magnetic realization, whose Sobolev norm uses covariant derivatives. This note distinguishes these domain prescriptions by an explicit graph limit for the ordinary restriction.

The construction places increasingly oscillatory functions at large x. Their y frequency cancels the vector potential near the center, making the graph norm grow more slowly than the ordinary y derivative. A disjoint sum then converges in graph norm while leaving ordinary H^1.

1 The ordinary Sobolev restriction

Let $\Omega = \{(x, y) \in \mathbb{R}^2 : y > 0\}$ and, for a fixed $c \in \mathbb{R}$, write

$$\mathcal{A}_c = -\partial_x^2 - \left(\partial_y - \frac{ix^2}{2}\right)^2 + icy. \tag{1}$$

All functions are complex-valued. We distinguish this differential expression from the operator T_c on $L^2(\Omega)$ with domain

$$D(T_c) = H_0^1(\Omega) \cap \{u \in L^2(\Omega) : \mathcal{A}_c u \in L^2(\Omega)\}, \qquad T_c u = \mathcal{A}_c u. \tag{2}$$

The expression in the domain is understood distributionally. Here H_0^1 is the closure of $C_c^\infty(\Omega)$ in the *ordinary* Sobolev norm, which includes both $\|\partial_x u\|_2$ and $\|\partial_y u\|_2$.

For a possibly nonclosed operator T on a Hilbert space H, let $\rho(T)$ consist of the numbers $z \in \mathbb{C}$ for which $T - z : D(T) \to H$ is bijective and its inverse is bounded as a map from H to H. We then set

$$\sigma(T) = \mathbb{C} \setminus \rho(T). \tag{3}$$

If spectrum is defined only for closed operators, the nonclosedness assertion below says instead that (2) does not meet that definition's prerequisite.

Proposition 1. *For every $c \in \mathbb{R}$, the operator T_c in (2) is densely defined and nonclosed. With convention (3), $\rho(T_c) = \varnothing$ and $\sigma(T_c) = \mathbb{C}$.*

Density follows from $C_c^\infty(\Omega) \subset D(T_c)$ and the density of test functions in L^2 [3, Corollary 4.23]. Every polynomial coefficient is bounded on the support of a test function, so the inclusion uses exactly the domain in (2). The proof of nonclosedness is given next; the resolvent consequence is established in Section 3.

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2 An explicit graph limit

Choose real functions $\phi \in C_c^\infty((-1/4, 1/4))$ and $\psi \in C_c^\infty((1,2))$ with $\|\phi\|_2 = \|\psi\|_2 = 1$. Norms of one-dimensional functions are taken on $\mathbb{R}$; all other norms in this section are on Ω. For $R \geq 1$ set

$$t = \sqrt{R}(x - R), \qquad k_R = \frac{R^2}{2}, \qquad v_R(x,y) = R^{1/4}\phi(t)\psi(y)e^{ik_R y}. \tag{4}$$

These are smooth compactly supported functions in Ω, and $\|v_R\|_2 = 1$ by the substitution $dx = R^{-1/2}dt$. The carrier frequency k_R cancels the vector potential at $x = R$. More precisely, on the support of v_R,

$$q_R(x) := k_R - \frac{x^2}{2} = -\sqrt{R}\,t - \frac{t^2}{2R}, \qquad |q_R(x)| \leq Q\sqrt{R}, \qquad Q = \frac{9}{32}. \tag{5}$$

Since q_R is independent of y, direct differentiation gives

$$\begin{aligned} \mathcal{A}_c v_R = R^{1/4} e^{ik_R y}\big[&- R\phi''(t)\psi(y) - \phi(t)\psi''(y) - 2iq_R(x)\phi(t)\psi'(y) \\ &+ q_R(x)^2\phi(t)\psi(y) + icy\phi(t)\psi(y)\big]. \end{aligned} \tag{6}$$

Thus

$$\|T_c v_R\|_2 \leq R\|\phi''\|_2 + \|\psi''\|_2 + 2Q\sqrt{R}\|\psi'\|_2 + Q^2 R + 2|c| \leq C_c R, \tag{7}$$

where one may take $C_c = \|\phi''\|_2 + \|\psi''\|_2 + 2Q\|\psi'\|_2 + Q^2 + 2|c|$. It is finite for every fixed c and independent of R. In contrast, the ordinary derivative has norm

$$\|\partial_y v_R\|_2^2 = \|\psi' + ik_R\psi\|_2^2 = \|\psi'\|_2^2 + \frac{R^4}{4}. \tag{8}$$

The last equality uses the reality of ψ and ψ', which makes the cross term zero.

Now let $R_n = 4^n$ and $a_n = R_n^{-3/2}$ for $n \geq 1$, and put

$$u_N = \sum_{n=1}^{N} a_n v_{R_n}, \qquad u = \sum_{n=1}^{\infty} a_n v_{R_n}, \qquad f_c = \sum_{n=1}^{\infty} a_n T_c v_{R_n}. \tag{9}$$

The x support of v_{R_n} lies in $I_n = (R_n - 1/(4\sqrt{R_n}), R_n + 1/(4\sqrt{R_n}))$. These intervals are pairwise disjoint: successive centers are separated by $3R_n$, whereas the sum of the two half-widths is $3/(8\sqrt{R_n})$. Derivatives and multiplication by the coefficients of $\mathcal{A}_c$ do not enlarge support, so the images in (9) are disjoint as well.

Every compact subset of Ω meets only finitely many of these supports. The two infinite sums therefore define actual locally smooth functions. Disjointness and (7) show that they belong to $L^2(\Omega)$, with the explicit tails

$$\|u - u_N\|_2^2 = \sum_{n>N} 4^{-3n} = \frac{4^{-3(N+1)}}{1 - 4^{-3}} \longrightarrow 0, \tag{10}$$

$$\|f_c - T_c u_N\|_2^2 \leq C_c^2 \sum_{n>N} 4^{-n} = \frac{C_c^2\, 4^{-(N+1)}}{1 - 4^{-1}} \longrightarrow 0. \tag{11}$$

The squared-norm identities for the infinite sums follow from monotone convergence applied to the nonnegative disjoint terms. Local finiteness also gives $\mathcal{A}_c u = f_c$ distributionally: a compactly supported test function sees only finitely many terms of (9).

Nevertheless $u \notin H^1(\Omega)$. Indeed, if $g \in L^2(\Omega)$ were its ordinary weak y derivative, then on the open strip $I_n \times (0,\infty)$ the weak derivative identity would give $g = a_n \partial_y v_{R_n}$ almost everywhere. Uniqueness here is the usual uniqueness of a locally integrable distribution [3, Corollary 4.24]. For every $M \geq 1$, (8) would then imply

$$\|g\|_2^2 \geq \sum_{n=1}^{M} a_n^2 \|\partial_y v_{R_n}\|_2^2 \geq \frac{1}{4}\sum_{n=1}^{M} R_n \geq \frac{4^M}{4}, \tag{12}$$

a contradiction. Thus $u \notin D(T_c)$, although $u_N \to u$ and $T_c u_N \to f_c$ strongly in L^2. Since every u_N belongs to $C_c^\infty(\Omega) \subset D(T_c)$, this proves that the graph is not closed. The functions are supported in $1 < y < 2$; the failure occurs at spatial infinity, away from the boundary.

3 The resolvent and the closed magnetic realization

For completeness, a bounded everywhere-defined inverse of a bijective shift forces an operator to be closed. To see this, suppose $B = (T - z)^{-1} : H \to H$ has these properties, and take $w_j \in D(T)$ with $w_j \to w$ and $Tw_j \to h$ in H. Boundedness gives

$$w_j = B(Tw_j - zw_j) \longrightarrow B(h - zw).$$

Hence $w = B(h - zw) \in D(T)$ and $Tw = h$. Applied to the nonclosed T_c, this proves $\rho(T_c) = \varnothing$ under (3), completing the proof of Proposition 1. The statement $\sigma(T_c) = \mathbb{C}$ does not assert that every complex number is an eigenvalue.

The domain distinction is important for interpreting this conclusion. The closed realization discussed in [2, pp. 6 and 12, equations (2.3), (2.4), and (2.24)] is defined using the magnetic space $H^1_{0,A}(\Omega)$, the closure of $C_c^\infty(\Omega)$ for the norm

$$\left(\|u\|_2^2 + \|\partial_x u\|_2^2 + \left\| \left(\partial_y - \frac{ix^2}{2} \right) u \right\|_2^2 \right)^{1/2}.$$

For $c \neq 0$, its form domain also requires $y^{1/2} u \in L^2(\Omega)$; the operator domain is then obtained by requiring the distributional expression to lie in L^2. This prescription uses the covariant derivative rather than the ordinary ∂_y norm in (8).

The result above concerns only the ordinary restriction (2), which is nonclosed. It identifies a defect in that printed domain, or gives the convention-dependent spectrum stated in Proposition 1. It provides no conclusion about spectral nonemptiness, eigenvalues, or a spectral threshold for the distinct natural closed magnetic Dirichlet realization. That all-parameter spectral question is left unanswered by this note.

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References

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